

Faderpunk WIP Apps — UX Cheatsheet

Synced to test/playground (IDs 29–46) · 16 Aug 2026 · A4 print

Main — Fader alone	Alt — Shift + Fader	Third — Channel button held + Fader
Short — Tap button	Long — Hold button (long-press)	Shift+... — Global Shift + gesture

Jack CV Out / CV In (Heat Pump, Grooves, Golden Gate, Arp de Levy, Chord Vamp, Echolot, Loop de Cay; Super LFO is 2-ch with CV on Ch0): **CV Out** = app drives the jack · **CV In** = jack modulates a **CV Dest** (attenuated by **CV Att** 0–100% where present). Device-cycle Dest: Heat Pump & Super LFO via **Shift+Short** (Grooves: Configurator only). Golden Gate + Jack=CV In → MIDI-first (no CV/gate outs).

ID 29Heat Pumpready

Clock-synced sidechain ducking envelope

FADERS

- Main** — Release (up = faster recovery)
- Alt** — Duck depth (red LED)
- Third** — Clock division (1/1 ... 1/32)

JACK PARAMS

- Jack: CV Out / CV In
- CV Dest: Depth · Release · Duck
- Dest colors: Red · Yellow · Cyan
- CV Att 0–100%

BUTTONS

- Short** — Manual duck trigger
- Long** — Mute (freeze level; no snap to idle; dim red button)
- Shift+Short (CV Out)** — Cycle clock division
- Shift+Short (CV In)** — Cycle CV Dest (flash)
- Shift+Long** — Invert duck direction

LEDS / OTHER PARAMS

- Button metronome Mid→Low per division · Mute = dim red (not off)
- 1/1 uses Color param
- Invert: white ↔ off fade
- Color, Range, MIDI CH/CC, NRPN, MIDI Out

Manual. Clock-synced ducking envelope: each division pulse ducks, then recovers by Release. Depth sets how far. Mute freezes the envelope at its current level (clock triggers discarded). Invert flips duck direction. Third sets division directly; Shift+Short cycles division (CV Out) or CV Dest (CV In). CV Att scales modulation.

ID 30Groovesready

Multi-genre MIDI drum grooves with feel and drummer personas

FADERS

- Main** — Density (extra hits)
- Alt** — Genre scrub (shared axis with Chord Vamp)
- Third** — Feel (capped by Swing max %)

GENRES

- Dub → Disco → House → Techno → Trip-Hop → Hip-Hop → Jungle → UK Garage → Dubstep
- LEDs: continuous open red→yellow→green→cyan→blue spectrum (same as Chord Vamp)
- Sound morphs continuously between neighboring genres along the axis

DRUMMERS

- Elvin → Purdie → Clyde → Jaki → Sly → Allen → Pretty Tony → Dilla
- Long press cycles; the button flashes a distinct colour per drummer
- The persona reshapes the genre pattern after it is read: shuffle, ghost density, backbeat lock, tom dispersal, clave rotation, roll rate, push or pull, limb drift and a machine amount
- All of it rides on Feel — at Feel 0 every drummer plays the plain genre grid, and the personality opens up as Feel rises
- Sly sits on the same index as Bassment's Robbie, so the pair lines up across both apps

BUTTONS

- Short** — Mute (notes off); unmuting also resets to the downbeat
- Shift+Tap** — Genre fill until next downbeat (break at high density)
- Shift+Hold** — Drum solo while held: 4-bar phrase (build → busier → peak → turnaround); resolves on release
- Long** — Next drummer (same gesture as Bassment's bassist)

JACK PARAMS

- Jack: CV Out Any 0–10V · CV Out Any -5–5V · CV Out Stacked 0–10V · CV Out Stacked -5–5V · CV In Density · CV In Feel · CV In Reset
- Duty and range share one param, so the sixteenth slot is free for the Drummer; stored layouts keep the duty and range they had
- Stacked sums families — hats 1, snare 2, kick 4 of 10 units; Open Hat counts as hats, Clap as snare, toms as kick
- The jack follows the groove, so it still pulses with every note switched off
- CV Att

PARAMS / LEDES

- Notes: Kick, Snare, Hats, Open Hat, Low Tom, High Tom, Clap — note 0 switches a voice off
- MIDI Ch is the base for all seven; Ch Map overrides it per voice (7 packed nibbles, 0 = follow base)
- Groove, Drummer, Swing max % (negative = reverse swing), GATE %, MIDI Out
- Groove = pocket: 0 is dead on the grid; up adds per-family ms drift (kick tight → hats loosest), velocity bow and walking ghosts
- GATE % is per voice — closed hats choke inside their step, open hats and toms may ring past it; CV pulse stays inside the 16th
- Button** Main: genre hue sat half→full (density) + bar lightness; hit dips kick -50% / snare -30% / hats -10%
- Button** Third: White→Color→Red (swing); same voice-weighted hit dips
- Button** Alt: committed genre hue (Top/Bottom = live fader meters; color changes only after latch pickup)

Manual. Genre MIDI drums on the global clock, sharing its nine genres with Chord Vamp and Bassment; the genre scrub morphs continuously between neighbours instead of stepping. Groove is the pocket: at zero every voice sits exactly on the grid, so Grooves locks tight against other Faderpunk apps. Turning it up drifts each family by its own few milliseconds — kick tightest, hats widest, scaled by the genre — bows velocity across the bar and walks which ghost hits Density reveals, so repeated bars stop reading identically. At high Density, sparse doubles, flams, triplets and rolls layer onto written kick/snare/hat hits (kick stays conservative; hats genre-typical); sub-strokes land inside the same 16th after swing/microtiming, never re-applying global swing. Shift+Tap plays a genre fill that crescendos toward the bar line independent of Density; above roughly 60 % Density it turns subtractive and the floor drops out for a break, handed back on the one by a late snare pickup. Shift+Hold escalates into a four-bar solo — build, busier bar, peak, turnaround — with accents pinned to the one and the backbeat, so Feel adds jitter instead of randomising the dynamics. Each genre carries three fill, break and solo figures; low Feel locks to one, higher Feel rolls between them. Clap, Open Hat and the toms are optional layers derived from hits already in the groove and stay silent until you give them a note: Open Hat replaces an accented offbeat hat rather than doubling it, and the toms only appear in the back half of a fill, break or solo. On top of the genre sits a drummer: eight personas that reshape the written pattern the way Bassment's bassists reshape a bassline, cycled with a long press. Purdie pulls the grid into a halftime shuffle and fills the gaps with ghosts while the backbeat stays nailed down; Jaki plays it dead straight and quantised; Pretty Tony rotates the kick off the downbeat into a clave and tumbles the toms; Dilla drifts the hats against a kick that sits behind the beat. Elvin, Clyde, Sly and Allen sit between those poles with their own mix of triplets, sixteenth chatter, steppers weight and afrobeat scatter. Because every persona trait is scaled by Feel, a flat Feel still gives the plain genre grid no matter who is behind the kit, and the character arrives as you open Feel up — so the drummer is a second dimension on the same fader, not a preset switch.

FADERS

- **Main** — Fibonacci depth
- **Alt** — Cycle length N
- **Third** — Local speed grid (16th → quarter; center = follow Speed param)

JACK PARAMS

- Jack: CV Out / CV In (In = MIDI-first, no outs)
- CV Dest: Depth · Cycle · Reset
- Range, CV Att

BUTTONS

- **Short** — Reset to downbeat
- **Long** — Mute
- **Shift+Short** — Reverse gap direction
- **Shift+Long** — Cycle Mode (Note/CC/Pitch/Phi)

MODE PARAM

- Note · CC · Pitch (12-TET) · Phi (ϕ pitch)
- Button color: default / orange / red / pink

PARAMS

- MIDI CH/Note/CC, GATE %, Speed, Color, MIDI Out

Manual. Fills cycle length N with Fibonacci gaps; deeper sets approach ϕ . Mode param selects Note/CC/Pitch/Phi; Shift+Long cycles the same on-device. Third overrides Speed locally. Short resets; Shift+Short reverses travel. Jack=CV In drops CV/gate outs and uses Dest for Depth/Cycle/Reset.

FADERS

- **Ch0 Main** — Morph · **Alt** — Attenuation (Amp)
- **Ch0 Third** — Skew
- **Ch1 Main** — Symmetry · **Alt** — Speed
- **Ch1 Third** — Phase-Warp

CV DESTINATIONS (CH0 SHIFT+SHORT)

- Blue Rate Mod · Pink Phase · Red Reset · Cyan Amp
- Yellow Speed · Orange Morph · Violet Skew
- Green Warp · Rose Symmetry

BUTTONS

- **Ch0 Short** — Freeze toggle (white)
- **Ch0 Long** — In mute
- **Ch0 Shift+Short** — Cycle CV Dest
- **Ch0 Shift+Long** — Cycle Mix Mode
- **Ch1 Short** — Phase reset
- **Ch1 Long** — Out mute
- **Ch1 Shift+Short** — Phase reverse
- **Ch1 Shift+Long** — Clock sync toggle

PARAMS

- Speed, Range, MIDI, Grid Lock
- Mix Mode: Xfade / Min / Max / Sum
- Osc B: Quad / Octave · Mix balance 0–100%
- Rate-mod depth 0–100% (Configurator) · CV Dest can modulate it
- Ch0 jack = CV In · Ch1 jack = CV Out

Manual. Dual-osc morphing LFO. Ch0 Morph/Amp/Skew on CV In; Ch1 Symmetry/Speed/Warp on CV Out. Shift+Short on Ch0 cycles CV Dest. Freeze holds the wave; In-mute silences CV-in; Out-mute silences output. Ch1 Shift+Long toggles clock sync (Grid Lock). Mix balance 0–100% (Xfade only).

FADERS

- **Main** — Delay (up = shorter / faster)
- **Alt** — Feedback
- **Third** — Pitch interval (± 24 st)

BUTTONS

- **Short** — Mute / ring-out
- **Long** — Panic (clear queue); skipped if Third used this hold

LEDS

- White flash on MIDI/gate in (~20% when muted)
- Ping-Pong: Top = Out A · Bottom = Out B

PARAMS

- I/O: MIDI→MIDI · MIDI→CV · CV→MIDI
- Delay: ms / Clock · Max delay
- Interval mode: Fixed / Stack / Bounce
- Routing: Single / Ping-Pong
- Signal: Pitch / Gate / CV→CC / Gate→Note
- MIDI In/Out A, Out Pong, CC, Note, Range, Color

Manual. Delay with feedback and optional pitch intervals. I/O picks MIDI/CV routing; Ping-Pong alternates MIDI Out A/B (MIDI→MIDI and CV→MIDI). Mute = ring-out (block new input; queue plays out). Long Panic clears the trail — Third interval edits never panic that hold. MIDI In always flashes white for diagnosis when muted.

FADERS

- **Main** — Evolve / mutation (0 = frozen)
- **Alt** — Texture (density / phrase / swing feel)
- **Third** — Lévy α / flight character

JACK PARAMS

- Jack: CV Out / CV In
- CV Out source (Jack=Out): Pitch · Gate · Velocity
- CV Dest (Jack=In): Evolve · Texture · Reroll
- Range, CV Att · V/Oct · Bypass quantizer

BUTTONS

- **Short** — Reroll note pool (+ reset)
- **Long** — Mute
- **Shift+Short (CV Out)** — Cycle CV Out source
- **Shift+Short (CV In)** — Reverse playback
- **Shift+Long** — Cycle octave span 1→2→3→4

OCTAVE LED

- Blue · Cyan · Yellow · Red

LEDS

- **Button / Top** — latch hue + level: Main = Color/Evolve · Alt = Orange/Texture · Third = Violet/ α ; mute = off
- **Bottom** — Pitch height in Color while gated (note activity)

PARAMS

- MIDI CH, Base Note, Color, MIDI Out
- Follow device tonic (default on)
- CV pitch + MIDI (dual path)

Manual. Generative arp whose pool evolves with Lévy-flight mutations. Main bottom freezes; raise to wander. Texture shapes phrase density/length; Third sets flight character (α). Shift+Long walks octave span (button color). Jack=CV Out: Shift+Short cycles Pitch/Gate/Velocity. Jack=CV In: Shift+Short reverses; CV modulates Evolve/Texture or fires Reroll. **Follow device tonic** (default on) makes the device Tonic a live transpose — hold the Scene button, move Fader 5, and the arp moves with Bassment, Contura, Chord Vamp and Venn. Because the pool holds absolute notes, it is *shifted* by the interval rather than rerolled, so the melodic shape it has evolved into survives the transpose. Takes effect at the next phrase start.

FADERS

- **Main (Perform)** — Scrub genre degrees × 3 octaves (pitch map)
- **Main (Auto)** — Tension (Meander only)
- **Alt** — Genre (+ Capture clip slot if clip exists)
- Shared axis: Dub → Disco → House → Techno → Trip-Hop → Hip-Hop → Jungle → UK Garage → Dubstep
- **Third (Auto)** — Feel (laid ↔ advanced stab length)
- **Third (Perform)** — Feel still stored; LEDs = blue / chord pitch

JACK PARAMS

- Jack: CV Out · CV In Macro · CV In Panic
- One enum — a CV In destination and CV Out were never live at once, and the freed slot is what lets Vamp grow
- Range, V/Oct, CV Att

BUTTONS

- **Perform hold** — Chord on; release = off (blue LEDs track pitch)
- **Auto short** — Play / pause
- **Shift+Short (Perform)** — Capture last N bars → Auto+Play
- **Shift+Short (Auto)** — Return to Perform
- **Auto Long** — Clear clip, reseed genre phrase
- **Shift+Long** — Panic (both modes)

PARAMS

- Root, Scale, Genre, Voicing, Velocity
- Follow device tonic (default on)
- Auto style: Repeat / Meander
- Start mode: Perform / Auto
- Capture: 16 / 8 / 4 / 2 / 1 bars
- LED: Perform = app color; Auto = orange; Hold Auto = rose (phrase pos)

SCENE STORAGE

- 8-slot vamp + Capture clip
- **Swing** 0–100 — genre-seeded; odd-16th delay (no Configurator param)
- Feel, scrub, tension, mode, play/pause

Manual. MIDI chord vamp with optional CV jack. Shared 9-genre axis with Grooves (Dub→...→Jungle→...→Dubstep). **Perform:** fader = genre-unique degrees ascending across ~3 octaves; hold plays the chord (blue = pitch). **Shift+Short** captures recent harmony (starts quantized to 16ths, lead-in cropped to first hit; hold lengths free) into a timed clip and arms Auto. **Auto:** genre rhythm phrase loops — multi-chord statement (~1–2 bars) then multi-bar rest; harmony Repeat or Meander; Feel sets short stab gate (rests fixed); Swing (scene, genre-seeded) delays odd 16ths on Auto + Capture. Short = play/pause; Long clears clip / reseeds; Shift+Short → Perform; Shift+Long = Panic in either mode. Alt = genre (+ Capture slot). CV Macro = scrub/tension; Panic from CV optional. **Follow device tonic** (default on) turns the device Tonic into a live transpose — hold the Scene button, move Fader 5, and the harmony moves with Bassment, Contura, Venn and Arp de Levy; the pitch class comes from the device, the octave stays with Root, and it lands on the next chord. There is no scale-follow counterpart: Scale defines the progressions, and Vamp sits at the 16-param ceiling.

FADERS

- **Main** — Gate % (Note) or CC amount (CC mode)
- **Alt** — Clock division
- **Third** — Velocity / sensitivity

BUTTONS

- **Short** — Play / trigger held sample
- **Shift+Short** — Manual sample (capture now)
- **Long** (no fader move) — Soft mute (fade; no hard note-off)

PARAMS / LEDS

- Mode Note / CC, MIDI CH/CC, Color, Range, V/Oct
- Layer meters: IR→UV spectrum on Main/Alt/Third fader
- White flash on each sample; mute = red button

Manual. Clocked sample-and-hold. Capture on division or Shift+Short; Main shapes gate length or CC depth. Long soft-mutes so notes can ring out. Third sets how strongly velocity follows the sample.

FADERS

- **Main** — Chord type (Unison · Power · Min · Maj · Sus4 · Min7 · Dom7 · Maj7 · Min9 · Add9)
- **Alt** — Inversion (0–2) / open (top + octave)
- **Third** — Velocity

BUTTONS

- **Short** (no fader move) — Mute / unmute all voices
- **Alt+Short** — Cycle harmony octave (–1 / 0 / +1)
- **Alt+Long** — Step voicing preset

PARAMS / LEDS

- I/O: MIDI→MIDI · MIDI→CV · CV→MIDI; MIDI In/Out CH
- CV Gate: Sustain · ms · clock divisions
- Strum (ms), Color, Range, V/Oct, Bypass quantizer
- Near-0V CV = rest; unpatched stays silent until stable pitch
- LEDs: IR→UV spectrum per layer; mute = red; preset flash on Alt+Long

Manual. One root (last MIDI note or pitch CV) picks the chord root; the fader picks chord type (Unison · Power · Min · Maj · Sus4 · Min7 · Dom7 · Maj7 · Min9 · Add9); up to four MIDI voices sound (MIDI→CV follows the highest). Alt sets inversion (0–2) or open top. Not a polyphonic keyboard input. Mute is a short tap — use Third (button+fader) for velocity so mute and vel don't collide. Alt+Short cycles harmony octave; Alt+Long walks voicing presets (tight/dry → wide/slow). CV Gate: Sustain, fixed ms, or clock divisions. Near-0V CV = rest; unpatched float stays silent until a stable pitch.

FADERS

- **Main** — Pitch (span from Base Note); spectrum LED meter. In Pitch→MIDI it transposes the CV pitch, so centre = unchanged and an unpatched jack leaves the fader in charge
- **Alt** (Shift+Fader) — Decay strength: bottom = immortal, top ≈ 4 wraps (violet→red)
- **Third** (Btn+Fader; gate modes while muted) — Decay mode zones (IR→UV): Random Rose · Pitch Up Orange · Pitch Down Yellow · Vel+Root Lime · Pitch Root Cyan · Velocity Blue · Gate Violet

BUTTONS

- **Hold** (gate modes) — Note on; release = off (long notes OK). Hold+Fader = scale-degree glissando toward the fader (like Chord Vamp)
- **Shift+Short** — Cycle loop length 1→2→...→8 bars (amber→cyan flash)
- **Shift+Long** — Mute (red LED; decay pauses)
- **Short while muted** — Unmute (re-arms)
- **Shift+Long while muted** — Erase buffer (white flash)

PARAMS / LEDS

- Mode: Pitch→MIDI · Gate→MIDI · MIDI→MIDI · MIDI→CV · Fader→Both
- Decay Mode: Random · Pitch Up · Pitch Down · Vel+Root (default) · Pitch Root · Velocity · Gate
- MIDI CH / In / Out, Base Note, Span (semitones across fader)
- Color, Range, V/Oct
- Capture/playback hit: sharp button 80% dim (~2 PPQN) + Top white; layer meters on Top/Bottom/Button

Manual. Clocked overdub loop: play while armed to capture; each wrap applies Alt strength through the selected Decay Mode (0 = freeze forever). Pitch modes stay on the global scale. Random jitters scale pitch and only lowers velocity. Gate shortens duration toward a 16th floor. Hold the button for long notes — mute is Shift+Long. Hold+Fader glissandos through scale degrees toward the fader (Chord Vamp-style). Length is Shift+Short (1–8 bars), not Alt/Third. Third picks Decay Mode (Configurator sync). Quantized to 16ths.

FADERS

- **Main** — Attenuverter –100...+100 % (center deadband = 0)
- **Alt** (Shift+Fader) — Offset –100...+100 %
- **Third** (hold button + fader) — Slew 0...10 s

BUTTONS

- **Short click** (no fader move) — Mute toggle on release (sends Output Min)
- **Hold + fader** — edit Slew (suppresses mute)
- **Shift+Hold** — momentary Bypass (keeps Range / Min-Max / Rate)

PARAMS / LEDS

- Pipeline: Gate/CV → Att → Offset → Transfer → Slew → Steps → Min/Max → Rate
- Button LED only: Main = app color × out; Alt = red × |offset|; Third = cyan × slew (floor so hue stays visible at 0)
- Mute = off · Bypass = full · black blink on downbeat while out > Min

Manual. 1-channel CV→MIDI sculptor. Defaults are transparent (+100 % att, 0 offset, 0 slew, Linear, full CC window, 100 Hz). Gate mode runs a Schmitt threshold (0.1 V hysteresis) then the same pipeline. Mute sends Output Min once and goes silent; unmute resends the live value. Watch the button LED for level — fader LEDs stay dark. Uses clock ticker only (no CLOCK_PUBSUB). No clock → no downbeat blink.

FADERS

- **Main** — F0 pulses A · F1 pulses B (density; absolute pulse count, Bjorklund redistributes)
- **Alt** (Shift) — F0 Extent (scale-degree span 0–24 from Note A) · F1 Interval (0–12 diatonic steps above the line for Ch2)
- **Third** (Btn1 or Btn2 hold) — F0 length A · F1 length B (2–32; independent)
- Rotation fixed at 0 (slots freed for Extent / Interval)
- **Division** (config) — note values 4/1 ... 1/32T (default 1/16)

BUTTONS

- **Btn1 or Btn2 hold** — Third (length A/B on F0/F1)
- **Btn2 tap** (no fader move) — Mute both outs
- **Shift+Btn1** — Cycle Logic OR → AND → XOR → Accnt
- **Shift+Btn2** — Toggle post-logic invert (shadow); white↔none fade like Heat Pump

MELODY / LEDS

- Note A (config) = degree 0 of the device quantizer key; Length-A walks an arch through 0...Extent scale degrees; Ch2 = same line + diatonic Interval
- Extent 0 = monotone Note A; higher = wider arch through the A cycle (device key)
- Division (note values), GATE %, Prob %, Humanize (Vel jitter; 0 = fixed max), MIDI CH / Note A / Interval B / Out, Color
- Color = App Color on Btn1; Btn0 = Logic hue; Top A/B = App Color / complement (layer hits)
- Main Btn: logic / app color · density bright · soft hit dip
- Alt: Top+Btn dim→sat ∝ Extent / Interval (fixed hues) · Third: shortest hue btn→complement ∝ length
- Top flashes = layer hits (Main) · Bottom flashes = logic outs (Main)
- Jack0 = logic result + melodic line · Jack1 coupled (OR→∩, AND→△, XOR→∩, Accnt→accent) + Interval

Manual. Two independent Euclidean masks combined with boolean logic — the OXI GEN dual-layer idea with a scale-aware melodic line. Color sets Layer A; Layer B LEDs use a fixed complement. Btn1 shows logic mode, not Color. Different lengths per layer enable polyrhythmic XOR/AND patterns Euclid cannot make alone. Melody: Note A is degree 0 of the device quantizer key. Extent is the arch span in scale degrees through Length-A; Interval B is diatonic steps above that line for Ch2 (wire indices 0–12 unchanged). Extent 0 = monotone Note A. Needs clock. Prob % < 100 adds optional flakiness; 100 is fully deterministic. No pitch CV (use GenSeq for that). Venn always draws its degrees from the device Scale. **Follow device tonic** (default on) adds the missing half: hold the Scene button and move Fader 5 and both lines transpose, together with Bassment, Contura, Chord Vamp and Arp de Levy. The pitch class comes from the device, the octave stays with MIDI Note A, and the change lands at the start of the next A-cycle rather than mid-pattern.

FADERS

- **Main** — Density (extra onsets + freer degrees)
- **Alt** — Genre scrub (shared axis with Grooves / Chord Vamp)
- **Third** — Feel / Groove (swing · microtiming · velocity · ghosts · gate; capped by Groove max %)

GENRES

- Dub → Disco → House → Techno → Trip-Hop → Hip-Hop → Jungle → UK Garage → Dubstep
- Alt scrub LEDs: continuous IR→UV spectrum (shared palette)

BUTTONS

- **Short** — Mute (notes off)
- **Shift+Tap** — Bass fill into next bar's chord (pedal break at high Density); white button LED
- **Shift+Hold** — Bass solo (+1–2 oct), re-rolls each bar; resolves on the next downbeat
- **Long** — Next Voice (flash = voice color)
- Previous Voice: Configurator only

JACK PARAMS

- Jack: CV Out (pitch V/Oct) · CV In Density · CV In Feel · CV In Reset
- One enum — a CV In destination and CV Out were never live at once
- Range, V/Oct, CV Att
- Filter CC — cutoff number (0 = off; 74 is typical); brightness = genre base + Voice bite + accent dynamics, lifted by fill / solo
- Groove max % carries the swing direction in its sign — negative swings the offbeats early
- High Density + Feel: genre ornaments (≤ 3 strokes/onset) — Voice shapes repeat, approach, ghost or octave pickup; filter CC only on the main strike

VOICES / LEDS

- Order = historical era: bebop → Motown → P-Funk → fusion → dub duo → alt-funk → quirk
- **Mingus** Violet — chromatic neighbors
- **Jamerson** Orange — pocket root/fifth
- **Bootsy** Pink — funk ghosts + octave drops
- **Jaco** Cyan — melodic approaches + leaps
- **Robbie** Green — dub pocket, sustained / sparse
- **Flabba** Lime — roots walk, roomy gates
- **Flea** Red — staccato + aggressive jumps
- **Claypool** Yellow — quirky leaps + chromatic nudges
- Button: genre hue in play; Long flash = voice color; fill/solo = white (dim = pedal break); hits duck Mid→Low

Manual. Genre MIDI bass on the global clock, sharing its nine genres with Grooves and Chord Vamp. Every sequenced hit retriggers with its own velocity and gate — including repeated pitches — so ornaments and ghosts stay audible. At high Density and Feel, genre-weighted doubles, flams, triplets and rolls add up to two monophonic sub-strokes per written onset (repeat, approach, ghost or octave pickup by Voice); fills and solos lean harder, filter CC coalesces on the main strike only. Phrases run eight bars in AA B A' form: the motif repeats while the harmony walks the genre's tropes, bars 5–6 answer, bar 7 ornaments, bar 8 breathes. At low Feel the line stays locked; raising it moves approaches, ghosts and octave punches from bar to bar — pickups appear before downbeats, soft notes drop out for air, ghosts turn into dead notes. Pocket Voices stay sparse in Jungle and UK Garage, busy ones thin out in Dub and Techno. Shift+Tap walks a fill toward the next bar's chord and lands its leading tone on the downbeat, the Voice picking the contour; at high Density it becomes a quiet held-root break. Shift+Hold lifts the register for a solo that re-rolls every bar while held and resolves on the next downbeat. Follow device tonic (on by default) turns the device Tonic into a live transpose, so Bassment changes key with Contura, Chord Vamp, Venn and Arp de Levy on the next bar line while keeping Root's octave. Follow device scale does the same for Scale and ships off, because the scale is part of the genre character.

FADERS

- **Main** — interval width (stepwise ↔ 2–3 octave leaps across the pool)
- **Alt** (Shift) — phrase length only (3–28 steps)
- **Third** (Btn hold) — note density (sparse rests ↔ continuous; audible mid)

BUTTONS

- **Short** — mute / unmute
- **Shift+Short** — octave span 1→2→3→4
- **Long** — next scale set
- **Shift+Long** — previous scale set

SCALE SETS / LEDS

- Flat list of named 12-TET sets (Ionian...Folk) — conventional interval-pattern labels, not cultural claims
- Follow device tonic (default on); Follow device scale uses the global Key mask instead of the selected set
- Per-set melodic feel (contour / leap / sustain / ornament) — expressivity is not a fader
- Ornaments = short grace, turn or repeat from feel + phrase position — not genre clichés; off when Division is too fine
- Button hue cycles by scale index · Top = gate · Bottom = pitch height · Division via config (default 1/8)
- Jack = CV Out (pitch) · CV In Density · CV In Interval · CV In Reset · CV In Phrase — one enum, since a CV In destination and CV Out were never live at once; CV Att scales input

Manual. Generates melodic contour phrases over selectable 12-TET pitch-class sets, anchored to the device quantizer tonic. Phrase starts and endings pull harder toward the tonic (cadence grammar); mid-phrase follows each set's ScaleFeel. Scale feel also drives brief ornaments — grace neighbours, turns or echoes — when the division leaves enough MIDI space (disabled on the finest grids). Scale-set names are conventional labels for interval patterns (Western modes and other named collections on equal footing). They are not claims about living musical practice or authentic intonation. Needs a running clock. Jack = CV Out (pitch, 1V/oct) or CV In (Density / Interval / Reset / Phrase + CV Att), set in the Configurator. CV on Phrase stretches and shortens the sentence, which moves the contour arch and the cadence with it. Gates via MIDI note on/off. Phrases mix short, medium, and longer note lengths. **Follow device tonic** (default on) turns the device Tonic into a live transpose: hold the Scene button and move Fader 5 and Contura moves with Bassment, Chord Vamp, Venn and Arp de Levy. The note pool is rebuilt on the change, so the cadence resolves in the new key; it takes effect at the next phrase start. **Follow device scale** (default off) swaps the Scale set for the device Scale.

FADERS

- **Main** — input trim on channel 0; attenuverter (CV) or threshold (Gate/Trigger) on channels 1–3
- **Alt** (Shift) — input offset on channel 0; output offset (CV) or hysteresis (Gate/Trigger) on channels 1–3
- **Third** (Btn hold) — input slew on channel 0; output slew (CV), minimum gate length (Gate) or pulse length (Trigger) on channels 1–3
- **Channel 0 while the internal LFO runs** — Main = LFO speed, Alt = morph, Third = skew (instead of trim, offset and slew)

BUTTONS

- **Channel 0 short** — freeze the conditioned input value
- **Channel 0 Shift + short** — while the internal LFO runs: free-running or clock-synced (the button flashes when sync engages)
- **Channels 1–3 short** — mute/unmute that output (Shift not held); fader scrub while the button is held cancels the tap
- **Channels 1–3 long** — cycle that out through CV → Gate → Trigger; the new mode hue appears immediately and the configurator follows
- **Channels 1–3 Shift + long** — cycle the output range ($\pm 5\text{ V} \leftrightarrow 0\text{--}10\text{ V}$); the button flashes **1x for $\pm 5\text{ V}$, 2x for $0\text{--}10\text{ V}$** ; stored as a parameter, so the configurator follows

SIGNAL / LEDS

- One CV input on channel 0, three outputs B/C/D on channels 1–3
- Five equidistant hues from blue through violet and red to orange, one per operating mode: channel 0 on CV in, channel 0 on the internal LFO, then CV, Gate and Trigger on the outs
- Channel 0 is blue while CV in is active and one step warmer when the internal LFO runs; each out takes the hue of its current mode, so a mode swap shows up immediately
- With nothing patched into the input, channel 0 becomes an internal morphing LFO that feeds B/C/D once the jack has been still for about 1.2 s; movement on the jack switches back to CV in
- *LFO Speed* shifts the whole speed range to Slow or Slowest
- CV mode uses the configured Range; Gate and Trigger modes drive the jack as a digital GPO and ignore Range
- Defaults: B = CV, C = Gate, D = Trigger ($\pm 5\text{ V}$ on the CV path)
- Channel 0 LEDs dim with the active source amplitude in the source hue
- CV outs dim with the shaped output level; Gate/Trigger meter the conditioned input while low and flash full when high
- Button LEDs use the same mode hues and breathe with their signal level: channel 0 stays bright when frozen; channels 1–3 turn off when muted
- MIDI: channel 0 and CV outs always send CC; Gate/Trigger outs can send Note instead via *Mode Map* (one bit per out) — pitch from that lane's *CC Map* slot; *Mode Map* 0 keeps all lanes on CC

Manual. One CV in, three shaped outs. Each out is independently CV, Gate or Trigger: CV gets attenuverter / offset / slew, Gate and Trigger get threshold / hysteresis / pulse length. A long press on channels 1–3 cycles that out through CV → Gate → Trigger right on the device, and the change is stored as a parameter, so the configurator stays in sync. Shift + long press on an out cycles its output range; the button flashes once for $\pm 5\text{ V}$ or twice for $0\text{--}10\text{ V}$ to confirm the new setting. Channel 0 freezes the conditioned input on a short tap; channels B/C/D mute on a short tap. With nothing patched into the input, channel 0 turns into a morphing LFO and drives B/C/D on its own once the jack has been still for about 1.2 s — movement on the jack switches back to CV in. Same wave engine as Super LFO, with speed, morph and skew on the three channel 0 fader layers (Main = speed, Alt = morph, Third = skew) and Shift + short press for clock sync. **LFO Speed** moves the whole rate range down to Slow or Slowest. All four waves — the conditioned input on channel 0 plus the three outs B, C and D — mirror to MIDI. Channel 0 and any out in CV mode always send CC. On Gate or Trigger outs, **Mode Map** can switch a lane to Note instead: one bit per out (B in bit 0, C in bit 1, D in bit 2); when set, that lane sends Note On/Off on edges and the pitch comes from that lane's **CC Map** slot (7-bit note number). **Mode Map** 0 keeps every lane on CC (legacy). By default they share the base **MIDI Channel** and take **MIDI CC** + 0...3. **Ch Map** and **CC Map** break that up: **Ch Map** packs one nibble per wave (wave 0 in bits 0–3, wave 1 in bits 4–7, ...), value + 1 = MIDI channel 1...16; **CC Map** packs 7 bits per wave (wave 0 in bits 0–6, wave 1 in bits 7–13, ...) holding the CC number directly (or the note number when **Mode Map** selects Note for that lane). A map of 0 means “all waves follow the base”, which is the shipped default and bit-for-bit the old behaviour. Example: channels 1, 5, 9 and 13 → nibbles 0, 4, 8, 12 → $0 + 4 \cdot 16 + 8 \cdot 256 + 12 \cdot 4096 = 51\,264$; CCs 20, 21, 40, 41 → $20 + 21 \cdot 128 + 40 \cdot 16384 + 41 \cdot 2097152 = 86\,641\,300$. A CC Map field is seven bits wide, so it reaches CC 0–127 only; in NRPN mode the base CC still spans the full 14-bit range, a mapped one does not. LEDs use five equidistant hues running from blue through violet and red to orange, one per operating mode. Channel 0 is blue while CV in feeds the outs and one step warmer once the internal LFO takes over; the outs take the remaining three steps by mode — CV, then Gate, then Trigger in orange — so a mode swap is visible the moment it lands. All four channels dim with signal amplitude.

FADERS

- **Main** — approach density in 7 steps: adds one approach chord at a time (V of each center, then ii of each center), growing the cycle from 3 to 9 chords on a fixed 9-slot grid
- **Alt** (Shift) — Feel: swing and velocity dynamics, bottom = straight and flat
- **Third** (Btn hold) — Time: picks the step division outright, 1/1 at the bottom down to 1/64t at the top, triplets interleaved

BUTTONS

- **Short tap** — mute (silences held chords, parks CV at 0 V)
- **Hold** — toggle reverse direction
- **Shift + tap** — reset cycle to step 0
- **Shift + hold** — cycle Interval: minor 3rd, major 3rd, 4th, tritone

SIGNAL / LEDS

- Three tonal center hues rotated around the app colour — top and bottom LEDs show the current center
- Button hue sways off the center hue with harmonic function: tonic on the center, its V a little away, its ii further
- Button brightness rises with Density across its seven steps, including the CV In Density offset
- CV Out tracks root pitch of the current step
- CV In Density modulates approach depth; CV In Reset triggers cycle reset on rising edge
- Button dips briefly on each chord trigger, scaled down from that density brightness
- Reverse fades the button white to none; Interval flashes one colour per step

Manual. Clock-driven chord sequencer cycling through three tonal centers spaced by a configurable interval (default major 3rd). Density fader controls approach depth in 7 positions on a fixed 9-slot grid — three slots (ii, V, I) per center. The I of each center always plays; raising the fader inserts one approach chord at a time, first the V of centers 1–3, then the ii of centers 1–3, so the cycle grows 3, 4, 5, 6, 7, 8, 9 chords. Empty slots aren't silent — the previous chord sustains through them, so the cycle stays nine slots long at every density and sparse settings give a long held tonic. Nine slots sit across 4/4, so the cycle returns to a downbeat every nine bars on binary divisions and every three on triplet ones. Chord qualities follow Coltrane convention: Maj7 on centers, Dom7 on dominants, m7 on subdominant approaches. Voicing param sets close triad, 7th, drop-2 or quartal stacking, and each chord is revoiced to move as little as possible from the one before it. Button tap mutes, cutting anything still held; hold toggles reverse. Shift + tap resets the cycle to step 0, shift + hold cycles the interval through minor 3rd, major 3rd, 4th and tritone. Direction param adds pendulum and random modes on top. CV out tracks the root pitch of the current step and parks at 0 V while muted; CV in modulates density or triggers reset. Shift + fader is Feel. At the bottom the sequence is straight and flat, exactly as before; further up the off-beats swing and the dynamics open out, accenting the tonic while the approach chords lighten. Feel steps aside when the device's own swing is active so the two never stack. Hold the button and move the fader for Time: it picks the step division outright across all eleven values, longest first — 1/1, 1/2, 1/4, 1/4t, 1/8, 1/8t, 1/16, 1/16t, 1/32, 1/32t, 1/64t. The Division param only sets where the app starts; the fader overrides it live. Motion decides how far the harmony is allowed to wander. Straight is the plain engine. Rubato lets the tonic lean, holding twice as long while the ii and V keep passing. Sheets breaks each chord into a fast run up and back down instead of a block. Free adds tritone substitutions on roughly every third dominant. Top and bottom LEDs show three distinct hues for the three tonal centers, rotated around the app colour. The button reads out the harmony itself: its hue sways off the current center hue with the harmonic function — the tonic sits on the center, its V a little away, its ii further still — while its brightness rises with Density across the seven steps, following the CV In Density offset when that jack mode is active. It dips briefly on each chord trigger, fades white to none when reverse toggles, and flashes a colour per step when the interval cycles.

FADERS

- **Main** — center position: three zones with snap; approach chords sound in the transition zones at boundaries
- **Alt** (Shift) — harmonic function: bottom = root (Maj7), middle = dominant (Dom7), top = subdominant approach (m7)
- **Third** (Btn hold) — Time: clock division from 1/1 at the bottom down to 1/64t at the top, triplets interleaved; the Division param sets the power-on starting point

BUTTONS

- **Short tap** — mute (silences held chords, parks CV at 0 V)
- **Hold** — toggle drone mode (sustain indefinitely, immediate fader response)
- **Shift + tap** — toggle latch (chord holds until tapped again)
- **Shift + hold** — cycle Feel in four steps, from straight and flat to full swing and dynamics

SUSTAIN

- **Clock-gated** — each chord holds until the next one
- **Legato** — same, and the last chord keeps ringing when the clock stops
- **Staccato** — every note releases halfway through its own step

SIGNAL / LEDS

- Three tonal center hues rotated around the app colour — brightness dims as fader drifts from center
- CV Out tracks root pitch of current position
- CV In Position modulates center selection; CV In Function modulates harmonic function
- Button blinks when latched, stays bright when droning, ducks on chord trigger
- Feel flashes one colour per step

Manual. Playable Coltrane tonal center navigator. Main fader positions across three centers with snap zones and approach-chord transitions at the boundaries. Alt fader selects the harmonic function at the current center: root (Maj7), dominant (Dom7) or subdominant approach (m7). Clock triggers the chord at the current grid position, and each chord is revoiced to move as little as possible from the one before it. Button tap mutes, cutting anything still held; hold enters drone mode, where the chord sustains indefinitely and updates immediately on fader movement, ignoring the clock. Shift + tap toggles latch (the chord sustains until tapped again regardless of fader movement), shift + hold cycles Feel through four steps. At the first step the playing is straight and flat; further up the off-beats swing and the dynamics open out, accenting the root chord while the approach chords lighten. Feel steps aside when the device's own swing is active so the two never stack. The Interval is set in the app params only and no longer cycles from the panel. Hold the button and move the fader for Time: the clock division, from 1/1 at the bottom down to 1/64t at the top, with the triplets sorted in by duration rather than tacked on at the end. The Division param is the power-on starting point — until the fader is touched for the first time it sits in the middle of that division's zone — and from then on the fader position is what counts and is remembered with the scene. Sustain sets how long the notes ring. Clock-gated holds each chord until the next one arrives. Legato does the same but also keeps the last chord ringing when the clock stops. Staccato releases every note halfway through its own step — half the chord's step for a block chord, half the note's slot in a Sheets or Free run. Motion decides how far the harmony is allowed to wander. Straight is the plain engine. Rubato lets the root chord lean, holding twice as long while the dominant and subdominant keep passing. Sheets breaks each chord into a fast run up and back down instead of a block. Free adds tritone substitutions on roughly every third dominant. Drone keeps its immediate feel and ignores the rhythmic side of Motion. LEDs show the current center hue, rotated around the app colour, and dim as the fader drifts from the center, visualising gravitational pull. The button blinks when latched, stays bright when droning, and flashes a colour per step when Feel cycles.

FADERS

- **Main** — root LFO speed on channel 0; modulation depth on channels 1–3
- **Alt** (Shift) — morph of the root wave on channel 0; base rate of the stage on channels 1–3
- **Third** (own button held) — skew of the root wave on channel 0; morph of the stage on channels 1–2
- **Channel 3 Third** — symmetry, shared by every stage instead of a private morph

BUTTONS

- **Channel 0 short** — freeze the root value; the whole chain stops moving
- **Channel 0 Shift + short** — clock sync for the root, available only while the internal LFO runs
- **Channels 1–3 short** — mute/unmute that stage; a fader scrub while the button is held cancels the tap
- **Channels 1–3 long** — cycle the modulation target through Rate → Depth → Shape; the channel hue follows and the button LED flashes
- The same three targets are also the params *Target B*, *Target C* and *Target D* in the configurator — the param sets the start value, a long press changes it live and pushes the new name back
- **Channels 1–3 Shift + long** — cycle the output range of that stage ($\pm 5\text{ V} \leftrightarrow 0\text{--}10\text{ V}$); the button flashes **1x for $\pm 5\text{ V}$, 2x for $0\text{--}10\text{ V}$** ; stored as a parameter, so the configurator follows

SIGNAL / LEDS

- Channel 0 is the root: a CV input that falls back to an internal LFO when the jack has been static for about a second
- Channel 1 is modulated by the root, channel 2 by the output of channel 1, channel 3 by the output of channel 2
- Every stage sees only its direct predecessor, so the chain diverges further towards the back
- *Rate* bends the tempo of the stage exponentially, roughly ± 3 octaves
- *Depth* is amplitude modulation — the stage ducks towards silence as its predecessor swings negative
- *Shape* pushes the wave of the stage along the morph axis
- *LFO Speed* shifts the root range to Slow or Slowest
- Top and bottom LEDs are a bipolar meter of the stage output
- The hue of channels 1–3 encodes the active modulation target; channel 0 changes hue depending on whether the CV input or the internal LFO drives it
- Button LEDs on channels 1–3 breathe with the stage output even while Shift is held, so the Alt-layer rate stays visible
- Muted channels are heavily dimmed and their button LED goes dark
- All four channels can mirror their level as MIDI CC or NRPN, using the base CC plus the channel offset

Manual. Ripppple is a chain of LFOs rather than four independent ones. Channel 0 is the root: its jack is a CV input, and when the signal has been static for about a second it falls back to an internal morphing LFO; jack movement switches back to CV in immediately. Channel 1 is modulated by the root, channel 2 by the output of channel 1, and channel 3 by the output of channel 2. Every stage only ever sees its direct predecessor, so what starts as a tidy relationship at the front of the chain diverges further and further towards the back — small moves on the root fan out into behaviour the last output can't easily be read back from. Each stage decides for itself what the incoming signal does to it. A long press on channels 1–3 cycles the modulation target through Rate, Depth and Shape: Rate bends the tempo of that stage exponentially over roughly ± 3 octaves, Depth is straight amplitude modulation that ducks the stage towards silence as its predecessor swings negative, and Shape pushes the wave along the morph axis. The same three targets are named in the configurator as the params “Target B”, “Target C” and “Target D” — the param sets the target a stage comes up with, and a long press on the device changes it live and pushes the new name back, so what you see on the screen and what the hardware does never drift apart. The faders carry three layers — the plain fader sets the root speed on channel 0 and the modulation depth on the stages, Shift reaches morph and base rate, and holding a channel's own button reaches skew, that stage's morph, and — on channel 3 only — the symmetry shared by the whole chain. Channel 0's button freezes the root value (the value itself, so stepped morph nodes hold too), Shift plus a tap puts the internal LFO on clock sync, a short tap on channels 1–3 mutes them with a click-free ramp, a long press cycles that stage's modulation target, and Shift plus a long press cycles its output range — the button flashes once for $\pm 5\text{ V}$ or twice for $0\text{--}10\text{ V}$ to confirm the new range. The LEDs meter the chain rather than merely decorate it. Top and bottom form a bipolar display of each stage's output, and the hue of channels 1–3 encodes the active modulation target, so a glance tells you which stage is being rate-bent and which is being ducked. Channel 0 switches hue depending on whether the CV input or the internal LFO is driving it — CV in when the jack moves, internal LFO after the idle timeout — so the automatic fallback is visible instead of mysterious. Muted channels drop to a heavy dim and their button LED goes dark; the button on channel 0 stays bright while the chain is frozen. On channels 1–3 the button LEDs keep pulsing with the stage output even while Shift is held, so the Alt-layer rate stays visible. All four channels can mirror their level as MIDI CC or NRPN, using the base CC plus the channel offset. Two packed params break that single-channel assumption when you need it. “Ch Map” holds one nibble per wave — wave 0 in bits 0–3, wave 1 in bits 4–7, and so on — where the stored nibble plus one is the MIDI channel 1–16. “CC Map” holds one 7-bit field per wave in the same order, wave w at bits $7\cdot w$ to $7\cdot w+6$, giving each wave its own CC number 0–127. A map of 0 as a whole means “follow the base”: Ch Map 0 sends every wave on the MIDI Channel param, CC Map 0 keeps the old base-CC-plus-offset derivation, and both default to 0, so an untouched Ripppple behaves exactly as before. For a worked example, to put the root on channel 1, stage 1 on channel 10 and the other two on channel 16, write the nibbles 0, 9, 15, 15, which is $0 + 9\cdot 16 + 15\cdot 256 + 15\cdot 4096 = 65\,424$. To send CCs 20, 21, 40 and 41, write $20 + 21\cdot 128 + 40\cdot 16384 + 41\cdot 2\,097\,152 = 86\,641\,300$. Note that a CC Map field is seven bits wide, so it can only reach CC 0–127 — in NRPN mode the base CC still spans the full 14-bit range, but a mapped one does not.